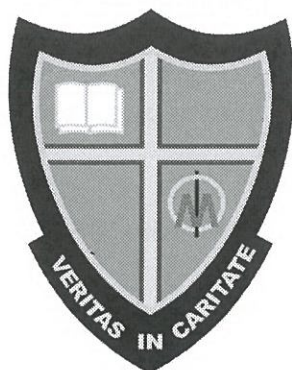


ST BENEDICT'S COLLEGE



To be checked
 $\approx$
 (Changed)

SUBJECT	APM Paper 1	DATE	23 July 2016
GRADE	11	MARKS	150
EXAMINER	Mrs Serafino	MODERATOR	Mrs Povall and Mrs Sillman
NAME		DURATION	2 hours
CLASS			

QUESTION	DESCRIPTION	POSSIBLE MARK	ACTUAL MARK
1	ALGEBRA	100	
2	CALCULUS	50	
	TOTAL	150	

STUDENT'S RESULT	
TEACHER'S SIGNATURE	
PARENT'S SIGNATURE	

QUESTION 1**21 MARKS**

No calculators may be used in this question. Show all working.

- a) Find the following in simplest surd form: $\sqrt{-12}$ 2
- b) Expand $(\sqrt{2008i^{2008}} - 1)^2$ and leave your answer in simplest surd form. 6
- c) Given: $f(x) = x^4 + 4x^3 + 3x^2 + 4x + 2$ with $f(-2 + \sqrt{2}) = 0$. Fully factorise $f(x)$ with $x \in \mathbb{C}$. 7
- d) $x = 2\sqrt{3}$ and $x = \sqrt{3}i$ are two solutions of the equation $x^4 + ax^2 + b = 0$. Find the values of a and b . 6

QUESTION 2**30 MARKS**Solve for x :

- a) $\frac{3}{2x^2-1} - \frac{1}{\sqrt{2x^2-1}} = 4$ 8
- b) $x\sqrt{x-2} - 5x - 3\sqrt{x-2} + 15 = 0$ 7
- c) $\frac{(x+3)^2(x-2)}{3-5x} \leq 0$ 6
- d) $\frac{-2}{x^2+x-20} \leq \frac{3}{x+5}$ 9

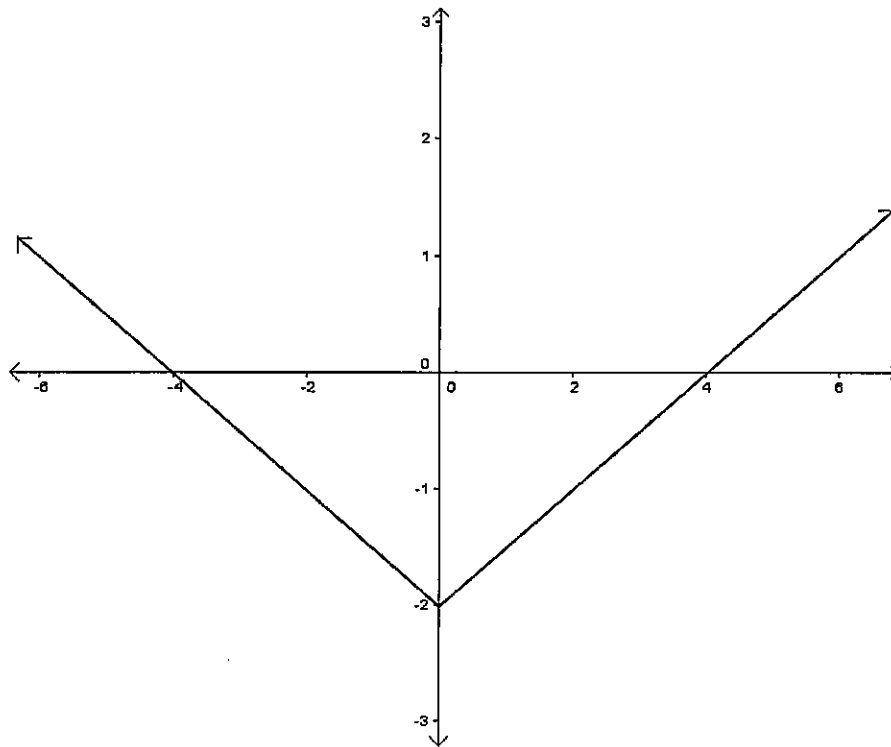
QUESTION 3**30 MARKS**

- a) Given: $f(x) = -2|x| + 2$ and $g(x) = \frac{1}{2}(2)^{x-1} - 1$
Sketch $f(x)$ and $g(x)$ on the same set of axes. Clearly show all the intercepts with the axes. 8
- b) Solve for x (with the aid of a diagram or otherwise):
- 1) $\left| \frac{1}{x} - 3 \right| > 3$ 8
- 2) $|x^2 - 3x - 4| < 2x - 3$ (Leave your answers in simplest surd form.) 14

QUESTION 4**13 MARKS**

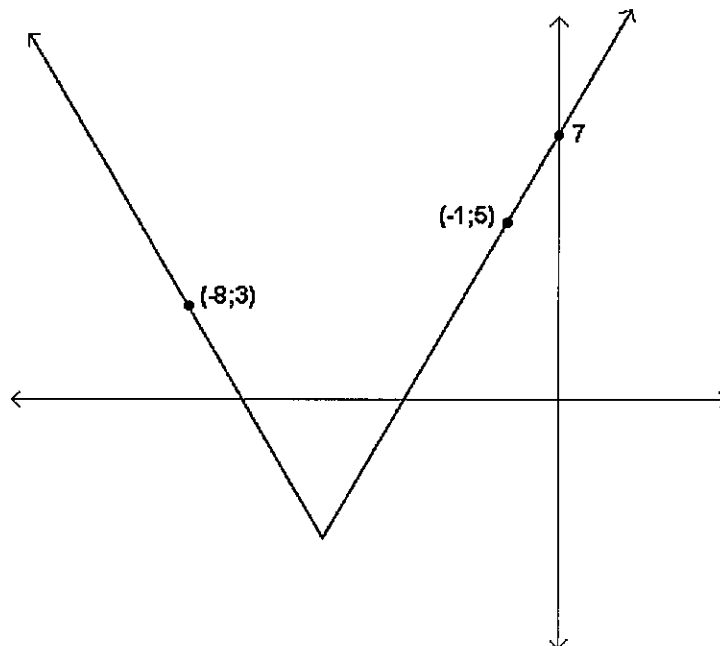
a) Determine the equations of the following graphs:

1)



4

2)



9

QUESTION 5**6 MARKS**Decompose into partial fractions: $\frac{2}{x^3+3x^2+2x}$

6

QUESTION 6**9 MARKS**

a) If $f(x) = x^2 - 3$, $g(x) = \sqrt{x} - 2$ and $h(x) = \frac{1}{x}$

1) Find without simplifying the expression of: $f \circ g \circ h(x)$. 3

2) Calculate the inverse of $g(x)$, write your answer as $g^{-1}(x) = \dots$ 2

b) Find the functions f and g such that $F = f(g(x))$ if:

1) $F(x) = \sin 5x$ 2

2) $F(x) = \sqrt{\frac{1}{x}} + 3$ 2

QUESTION 7**15 MARKS**

a) Sketch the graph of f if: 10

$$f(x) = \begin{cases} x+2 & \text{if } x < -1 \\ x^2 - 1 & \text{if } -1 \leq x < 2 \\ \frac{3}{2}|x-4| & \text{if } x > 2 \\ 4 & \text{if } x = 2 \end{cases}$$

b) Give the zeroes of $f(x)$. 3

c) When x gets very close to 2; what value does $f(x)$ get very close to? 1

d) What is $f(2)$ equal to? 1

Please ensure that your calculator is set in radians for the rest of this paper.

QUESTION 8**15 MARKS**

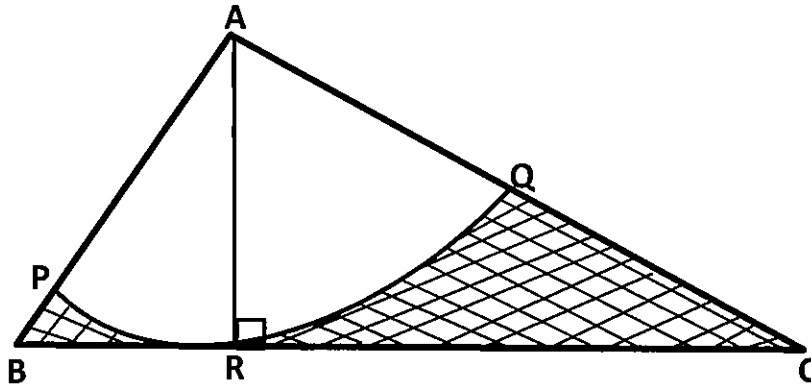
a) Simplify without the use of a calculator: $\cos\left(\frac{7\pi}{3}\right) \cot\left(\frac{7\pi}{4}\right)$ 4

b) Prove: $\frac{\cot^2 x - 1}{\cot^2 x + 1} = \frac{2}{\operatorname{cosec}^2\left(\frac{3\pi}{2} - x\right)} - 1$ 6

c) Determine the general solution for: $\frac{1}{2} \operatorname{cosec} 2\theta + 1 = 0$ 5

QUESTION 9**11 MARKS**

Triangle ABC is such that $AB = 5$ cm and $BC = 7$ cm. The point R is the foot of the perpendicular from A to BC such that $BR = 2$ cm. With centre A and radius AR, a circular arc is drawn, from a point P on AB to a point Q on AC, touching the line BC at R.



- | | |
|---|---|
| a) Find the length of AR, leaving your answer in surd form. | 2 |
| b) Find the size of angle $\widehat{BAC}$ in radians. | 5 |
| c) Find the area of the shaded region, which lies inside triangle ABC but outside sector APQ. | 4 |

Question 1

121

$$\begin{aligned} \text{a) } \sqrt{-12} &= \sqrt{12} \cdot \sqrt{-1} \\ &= 2\sqrt{3}i \quad \checkmark \quad (2) \end{aligned}$$

$$\begin{aligned} \text{b) } (\sqrt{2008} i^{2008} - 1)^2 &= 2008 i^{2008} - 2\sqrt{2008} i^{2008} + 1 \quad \checkmark \\ &= 2009 - 4\sqrt{502} \quad \checkmark \end{aligned}$$

$\left. \begin{aligned} i &= \sqrt{-1} \\ i^2 &= -1 \\ i^3 &= -i \\ i^4 &= 1 \end{aligned} \right\} \text{Pattern}$

$\therefore i^{2008} = 1 \quad \checkmark$
(6)

$$\begin{aligned} \text{c) } x &= -2 + \sqrt{2} \quad \therefore x = -2 - \sqrt{2} \quad \checkmark \\ \therefore x^2 + 4x + 2 & \quad \checkmark \quad (4 - 2 = 2) \end{aligned}$$

$$\begin{aligned} f(x) &= (x^2 + 4x + 2)(x^2 + bx + 1) \\ &= (x^2 + 4x + 2)(x^2 + 1) \quad \checkmark \\ &= (x + 2 - \sqrt{2})(x + 2 + \sqrt{2})(x + i)(x - i) \quad \checkmark \quad \checkmark \quad \checkmark \end{aligned}$$

$\begin{aligned} bx^3 + 4x^3 &= 4x^3 \\ b &= 0 \end{aligned}$
(7)

$$\begin{aligned} \text{d) } x &= 2\sqrt{3} \quad \therefore x = -2\sqrt{3} \quad \checkmark \\ \therefore x^2 - 12 & \\ x &= \sqrt{3}i \quad x = -\sqrt{3}i \quad \checkmark \\ x^2 + 3 & \quad \checkmark \\ (x^2 - 12)(x^2 + 3) &= 0 \\ x^4 - 9x^2 - 36 &= 0 \\ \therefore b &= -9 \quad \checkmark \quad a = 36 \quad \checkmark \quad (6) \end{aligned}$$

Question 2

30

a) let $k = \sqrt{2x^2 - 1}$ ✓

$\therefore k^2 = 2x^2 - 1$ ✓

$\frac{3}{k^2} - \frac{1}{k} = 4$

$3 - k = 4k^2$

$0 = 4k^2 + k - 3$ ✓

* $0 = (4k + 3)(k - 1)$ ✓

$\therefore k = -\frac{3}{4}$ or $k = 1$ ✓

Check soln!

$\sqrt{2x^2 - 1} = -\frac{3}{4}$ ✓ or $\sqrt{2x^2 - 1} = 1$ ✓

$2x^2 = \frac{9}{16}$ no sol.

not sol.

$2x^2 = 1 \Rightarrow 1 = 1$

$2x^2 = 2$

$x^2 = 1$

$x = \pm 1$ ✓ (8)

(b) $x\sqrt{x-2} - 5x - 3\sqrt{x-2} + 15 = 0$

$x(\sqrt{x-2} - 5) - 3(\sqrt{x-2} - 5) = 0$ ✓

$(\sqrt{x-2} - 5)(x - 3) = 0$ ✓

$\sqrt{x-2} = 5$ ✓ or $x = 3$ ✓

$x - 2 = 25$ ✓

$x = 27$ ✓

(7)

(c) $\frac{(x+3)^2(x-2)}{3-5x} \leq 0$

$\frac{(x+3)^2(x-2)}{(5x-3)} \geq 0$ ✓

CV's: $2; \frac{3}{5}$

$(x+3)^2 > 0$ ✓

$\begin{array}{c} + & 0 & - & 0 & + \\ & \frac{3}{5} & & 2 & \end{array}$ ✓

$\leftarrow 0 \quad 0 \quad 0 \rightarrow$ ✓

$x < \frac{3}{5}$ or $x \geq 2$ ✓

(6)

$$(d) \frac{-2}{(x+5)(x-4)} - \frac{3}{(x+5)} \leq 0$$

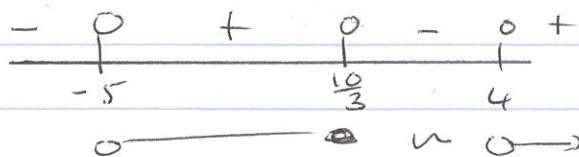
$$\frac{-2 - 3(x-4)}{(x+5)(x-4)} \leq 0$$

$$\frac{-2 - 3x + 12}{(x+5)(x-4)} \leq 0$$

$$\frac{3x - 10}{(x+5)(x-4)} \geq 0$$

$$\frac{3x - 10}{(x+5)(x-4)} \geq 0$$

$$CV's : \frac{10}{3}, -5, 4$$



✓✓ (any method)

$$-5 < x \leq \frac{10}{3} \text{ or } x > 4$$

(9)

Question 3

130

$$(a) f(x) = -2|x| + 2$$

salient pt: (0; 2)

Shape:

y-int: 2

$$x\text{-int} : 2|x| = 2$$

$$|x| = 1$$

$$\therefore x = \pm 1$$

$$y(x) = \frac{1}{2}(2)^{x-1} - 1$$

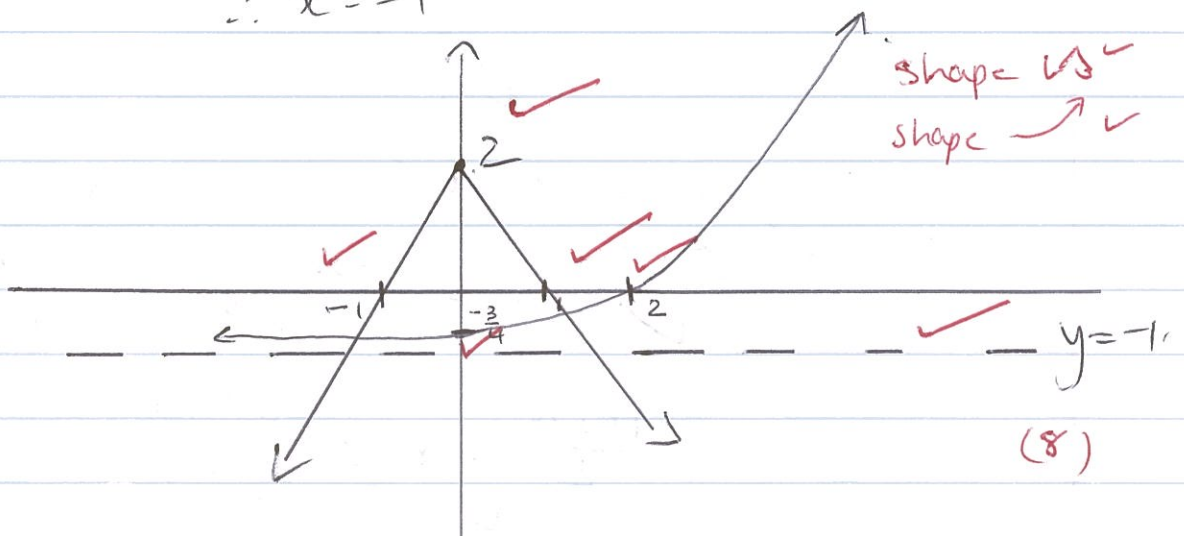
$$y\text{-int} : -\frac{3}{4}$$

$$x\text{-int} : 0 = \frac{1}{2}(2)^{x-1} - 1$$

$$1 = \frac{1}{2}(2)^{x-1}$$

$$2 = 2^{x-1}$$

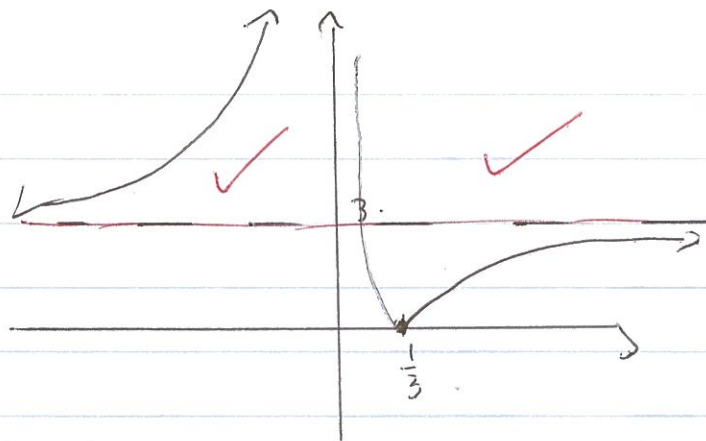
$$2 = x$$



shape ✓
shape ✓

(8)

b(1)



[Do not need a sketch]

$$\frac{1}{x} - 3 = 3$$

$$\frac{1}{x} = 6 \quad \checkmark$$

$$\therefore x = \frac{1}{6} \quad \checkmark$$

$$\therefore x < \frac{1}{6}; x \neq 0.$$

$$\text{and? } x < 0$$

(8)

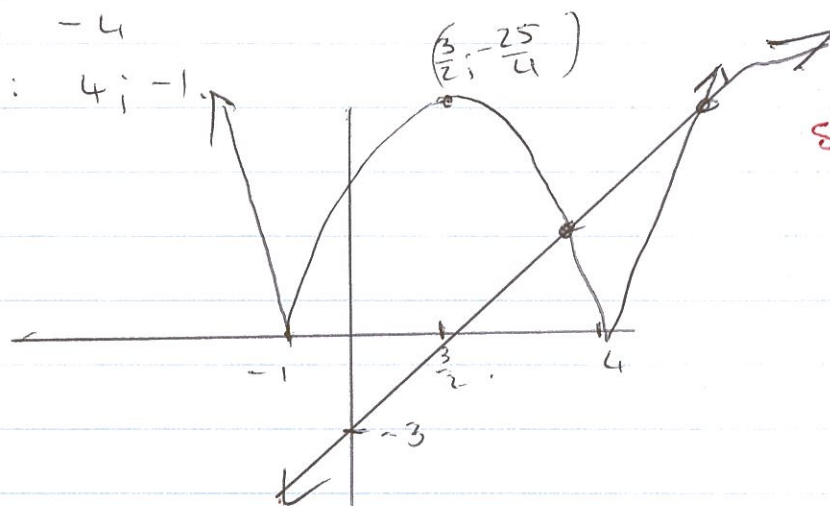
$$(2). |x^2 - 3x - 4| < 2x - 3$$

$$\text{tpc: } x = \frac{3}{2}$$

$$y = -\frac{25}{4}$$

$$y_{\text{int:}} -4$$

$$x_{\text{int:}} 4; -1$$



sketch: 4
2

$$-x^2 + 3x + 4 = 2x - 3 \quad \checkmark$$

$$0 = x^2 - x - 7$$

$$x = \frac{1 + \sqrt{29}}{2} \quad \checkmark \text{ or } x = \frac{1 - \sqrt{29}}{2} \quad \checkmark$$

$$\therefore \frac{1 + \sqrt{29}}{2} < x < \frac{5 + \sqrt{29}}{2}$$

$$x^2 - 3x - 4 = 2x - 3 \quad \checkmark$$

$$x^2 - 5x - 1 = 0$$

$$x = \frac{5 + \sqrt{29}}{2} \quad \checkmark \text{ or } x = \frac{5 - \sqrt{29}}{2} \quad \checkmark$$

(14)

Question 4:

$$a (1) \quad y = a|x-p| + q$$

$$y = a|x| - 2$$

$$0 = a|4| - 2$$

$$2 = 4a$$

$$\frac{1}{2} = a$$

$$\therefore y = \frac{1}{2}|x| - 2$$

$$(2) \quad y = mx + 7$$

$$5 = -m + 7$$

$$2 = m$$

$$\therefore y = 2x + 7$$

$$y = -2x + c$$

$$3 = -2(-8) + c$$

$$-13 = c$$

$$y = -2x - 13$$

$$\therefore 2x + 7 = -2x - 13$$

$$4x = -20$$

$$x = -5$$

$$\therefore y = -3$$

$$y = a|x+5| - 3$$

$$7 = a|0+5| - 3$$

$$10 = 5a$$

$$2 = a$$

$$\therefore y = 2|x+5| - 3$$

Question 5: 6

$$\frac{2}{x(x+2)(x+1)} \checkmark = \frac{A}{x} + \frac{B}{x+2} + \frac{C}{x+1} \checkmark$$

$$2 = A(x+2)(x+1) + B(x)(x+1) + C(x)(x+2)$$

$$x=0 \quad 2 = A(2)(1) \\ 1 = A$$

$$x=-1 \quad 2 = C(-1)(-1+2) \\ -2 = C$$

$$x=-2 \quad 2 = B(-2)(-2+1) \\ 1 = B$$

$$\therefore \frac{2}{x(x+2)(x+1)} = \frac{1}{x} - \frac{2}{x+1} + \frac{1}{x+2} \checkmark \checkmark \checkmark \quad (6)$$

Question 6: 9

$$\begin{aligned} \text{a) 1) } g \circ h(x) &= \sqrt{\frac{1}{x}} - 2 \checkmark \\ \therefore f \circ g \circ h(x) &= (\sqrt{\frac{1}{x}} - 2)^2 - 3 \checkmark \quad (3) \end{aligned}$$

$$\begin{aligned} \text{2) } x &= \sqrt{y} + 2 \\ x-2 &= \sqrt{y} \checkmark \\ (x-2)^2 &= y \\ \therefore g^{-1}(x) &= (x-2)^2 \checkmark \quad (2) \end{aligned}$$

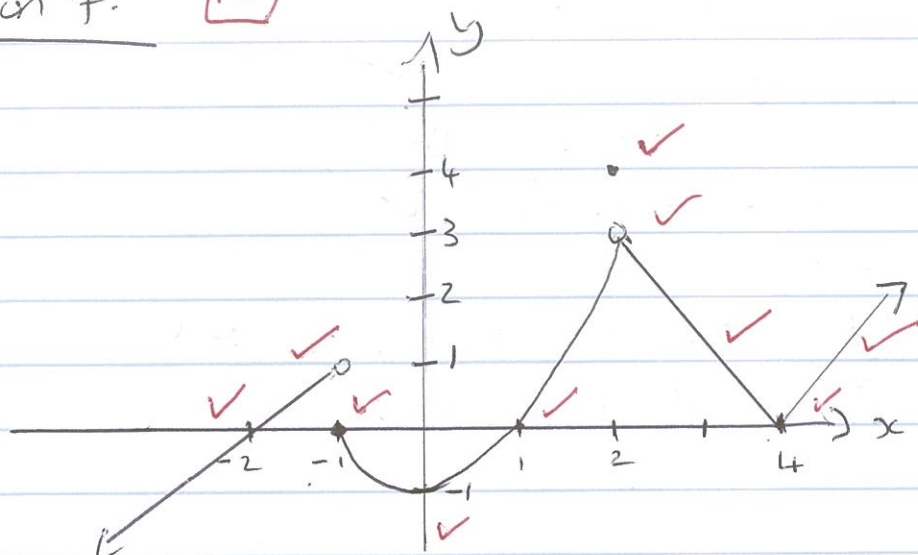
$$\begin{aligned} \text{b) 1) } f(x) &= \sin x \checkmark \\ g(x) &= 5x \checkmark \quad (2) \end{aligned}$$

$$\begin{aligned} \text{2) } f(x) &= \sqrt{x} + 3 \checkmark \\ g(x) &= \frac{1}{x} \checkmark \quad (2) \end{aligned}$$

Question 7:

15

(a)



(10)

(b) 1) $x = -2; -1; 1; 4$.

(3)

2) 3

(1)

(3) 4

(1)

Question 8:

15

$$\begin{aligned}
 (a) & \cos\left(\frac{7\pi}{3}\right) \cot\left(\frac{7\pi}{4}\right) \\
 &= \cos\left(2\pi + \frac{\pi}{3}\right) \cot\left(2\pi - \frac{\pi}{4}\right) \\
 &= \cos\frac{\pi}{3} \cdot \left(-\cot\frac{\pi}{4}\right) \\
 &= \frac{1}{2}(-1) \\
 &= -\frac{1}{2}
 \end{aligned}$$

(4)

$$(b) \frac{\cot^2 x - 1}{\cot^2 x + 1} = \frac{2}{\operatorname{cosec}^2\left(\frac{3\pi}{2} - x\right)} - 1$$

$$\begin{aligned}
 \text{LHS} &= \frac{\left(\frac{\cos^2 x}{\sin^2 x} - 1\right)}{\operatorname{cosec}^2 x} \\
 &= \frac{\cos^2 x - \sin^2 x}{\sin^2 x} \div \frac{1}{\sin^2 x} \\
 &= \cos^2 x - \sin^2 x
 \end{aligned}$$

$$\begin{aligned}
 \text{RHS} &= \frac{2}{\sec^2 x} - 1 \\
 &= 2\cos^2 x - 1 \\
 &= \cos^2 x - \sin^2 x
 \end{aligned}$$

$$\therefore \text{RHS} = \text{LHS} \quad (6)$$

$$(c) \quad \frac{1}{2} \operatorname{cosec} 2\theta + 1 = 0$$

$$\frac{1}{2 \sin 2\theta} = -1 \quad \checkmark$$

$$1 = -2 \sin 2\theta$$

$$\therefore -\frac{1}{2} = \sin 2\theta \quad \checkmark$$

$$\text{calc} = -\frac{\pi}{6}$$

$$2\theta = -\frac{\pi}{6} + 2\pi k; k \in \mathbb{Z} \quad \checkmark$$

$$\theta = -\frac{\pi}{12} + \pi k \quad \checkmark$$

or

$$2\theta = \pi - (-\frac{\pi}{6}) + 2\pi k; k \in \mathbb{Z} \quad \checkmark$$

$$2\theta = \frac{7\pi}{6} + 2\pi k$$

$$\theta = \frac{7\pi}{12} + \pi k \quad \checkmark \quad (5)$$

Question 9:

(11)

$$a) \quad s^2 = z^2 + AR^2$$

$$z^2 = AR^2 \quad \checkmark$$

$$\sqrt{z^2} = AR \quad \checkmark$$

(2)

$$(b) \quad AC^2 = (\sqrt{z^2})^2 + (s)^2$$

$$\therefore AC = \sqrt{46} \quad \checkmark$$

$$(7)^2 = (\sqrt{46})^2 + (\sqrt{z^2})^2 - 2(\sqrt{46})(\sqrt{z^2}) \cos \widehat{BAC} \quad \checkmark$$

$$-18 = -2\sqrt{966} \cos \widehat{BAC} \quad \checkmark$$

$$0,289 \dots = \cos \widehat{BAC}$$

$$1,277 = \widehat{BAC} \quad \checkmark$$

(5)

$$(c) \quad \frac{1}{2} (\sqrt{z^2})(7) - \frac{1}{2} (1,277)(\sqrt{z^2})^2 \quad \checkmark$$

$$= 2,63 \text{ cm}^2 \quad \checkmark$$

(4)